M.E. MODEL PROJECT REVIEW • PROVISIONAL STRUCTURE

AVA Evaluation

An Empirical Study of Conflict Resolution for Assertional State in Multi-Agent AI Systems



OWL ANALOGY

Agree · Verify · Abstain

Proposal/review aid — not an official KEC template, thesis chapter, or implementation report.

Eight provisional modules — a route through the proposal

01 Problem and background

Why shared meaning can fail even when replicas agree

02 Scope and state model

Separate additive facts from single-valued assertions

03 Relation classes

Classify the relationship before choosing a policy action

04 Agree / Verify / Abstain model

Keep policy actions distinct from terminal outcomes

05 Research questions and comparison arms

Compare strong baselines fairly, and separate classifier from policy

06 Dataset, annotation and metrics

Build a trustworthy gold set; report safety errors separately

07 SCP-MA experiment and convergence target

Use a minimal fault simulator only after the research gate

08 Risks, limitations and staged plan

Protect the research gate and keep the M.E. scope manageable

Replicas can agree while meaning stays unsafe

Agent A

database = PostgreSQL

Agent B

database = DynamoDB

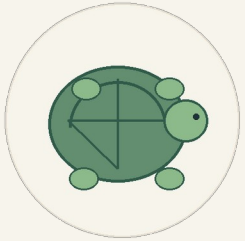
Both events reach every replica

What AVA asks

Are the claims paraphrases, compatible, soft conflict or hard conflict?

Should the system agree, verify or abstain?

Mechanical replication ≠ semantic correctness



TURTLE ANALOGY: pause before merging

Scope: additive state versus assertional state

A • Additive state

Logs, observations, discovered entities
Mergeable facts; standard G-Set / OR-Set CRDTs
Not claimed as novel

Q • Assertional state

Single-valued fact per declared key
Competing values may conflict
Primary research object



BEE ANALOGY: separate roles

Assertion q = (key, value, source agent, causal metadata, logical timestamp)

Same-key scope must be explicit: key type, role, temporal scope, and cardinality.

Out of scope: Byzantine agents · permanent partitions · cross-key intent conflicts · general open-domain NLI

Four relation labels — before policy actions

Paraphrase

Same fact, different wording

PostgreSQL is selected / Project uses PostgreSQL

Compatible

Both can be true in declared scopes

Transactions use PostgreSQL / Sessions use DynamoDB

Soft conflict

Different claims; policy/time may resolve

API v1 active / migrating to API v2

Hard conflict

Mutually exclusive for same key/time

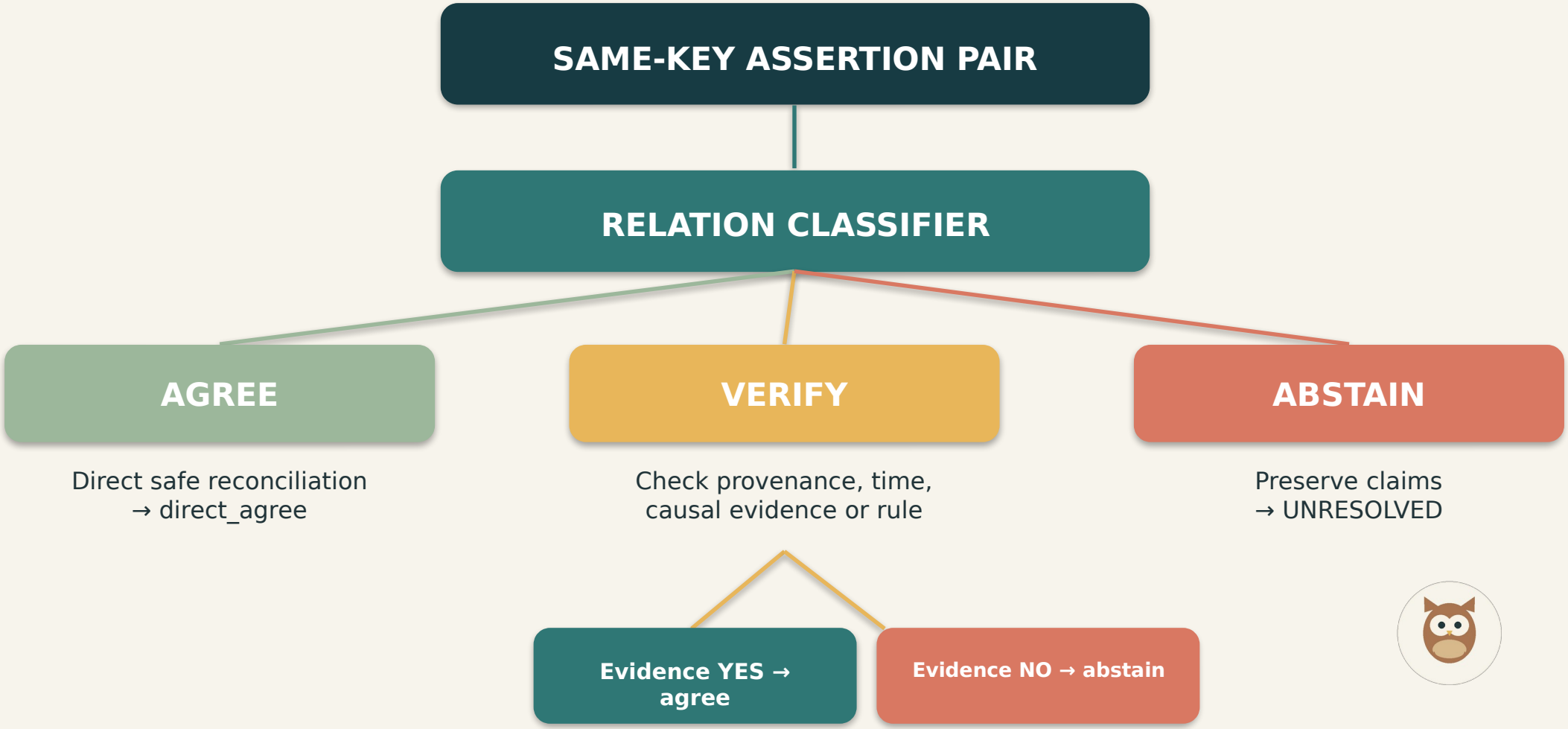
database=PostgreSQL / database=DynamoDB

Label the relationship first. Choose system action second.



FOX ANALOGY: notice subtle differences

AVA policy: Agree, Verify, or preserve as Unresolved



OWL ANALOGY

Seven arms — two different layers to compare

RELATION / CLASSIFICATION SIGNALS

B2 Embedding	similarity
B3 NLI	entail / contradict / neutral
B4 Hybrid SCS	weighted semantic signals
B5 LLM Judge	fixed prompt/model; repeated trials
B6 Jev	structured System-One questions

RESOLUTION / SAFETY POLICIES

B1 Deterministic	argmax(timestamp, agent_id)
B7 Preserve / abstain	non-identical remains unresolved
AVA Verify stage	evidence-based intermediate check

Fairness rule: report classifier accuracy separately from final safety and resolution outcomes.

Dataset: human-labelled pairs, context and evidence

1. **Controlled synthetic**
   balanced edge cases
2. **LLM-generated realistic**
   human checked
3. **Real traces (if feasible)**
   legal/practical access

Record fields

Declared key + type · assertion A/B · context/time
Evidence/provenance · gold label · difficulty

Gold-label process

3 annotators → agreement statistic → adjudication
LLM output alone is not ground truth



BEE ANALOGY

Safety metrics stay separate — no misleading composite

FMR • unsafe agreement

Wrong Agree on actual conflicts (direct or verified) ÷ actual conflicts

FCR • missed safe resolution

Failure to resolve actual Paraphrase + Compatible cases ÷ those resolvable cases

Unnecessary abstention

Subset of FCR after Verify on resolvable cases; reported separately too

Report a safety–coverage picture, not one “winner” score.

Also: per-class F1 · calibration (ECE / Brier) · latency · token / compute cost · throughput



TURTLE ANALOGY

SCP-MA: test decisions under asynchronous event delivery

Agents

concurrent writes

Immutable log

event IDs + causal metadata

Network

delay · duplicate · reorder · disconnect

Projection

resolve value or UNRESOLVED

duplicate delivery

reordered delivery

delayed delivery

Measure

Convergence rate/latency · divergence · bytes/messages · duplicate handling · recovery

First validation

Semantics-free deterministic simulator before semantic methods are connected



PENGUIN ANALOGY

Convergence is a target condition — not yet proved

Same valid event set E + deterministic Resolve(Canonicalize(E))
⇒ identical projected state at every replica

Canonicalize

same event set

Resolve

pure deterministic function

Project

same value or UNRESOLVED

Open proof obligations: canonical inputs · tie rules · deterministic Verify · exact meaning of “state”

Prior work narrows broad claims

Pair relations / false merges	**MELD**	104 pairs; distinct keys
Conflict-preserving state	**StateFuse · TOKI**	history, provenance, projection, abstention
Long-history relational QA	**SubtleMemory · TANGLE · STALE**	downstream memory tasks
Coordination / governance	**S-Bus · Governed Shared Memory**	structural concurrency / supersession
NLI / evidence pairs	**ECon · RefNLI · ANLI**	adjacent pair reasoning; different setup

Interpretation: substantial overlap exists; novelty remains unresolved pending full source-wide verification.



State the literature numbers with their caveats

MELD

104-pair set uses distinct claim keys

CodeCRDT

5–10% from manual 60/600 sample

SCF

27.9% precision · 65.2% recall · F1 39.0

Freshness pipeline

+10.8 / +21 pp are whole-pipeline; executor-only +2 / 0

Partial findings challenge scope; they are not a final novelty verdict.

Key decisions before any build

Key schema

Cardinality, type, role, temporal scope

Label guide

Soft vs hard; pair vs context

Layer boundary

Classifier output → policy action mapping

Evidence for Verify

Allowed deterministic sources and rules

Feasibility

Annotators, sample size, compute/API budget

Requirements

Supervisor, college format and milestones unknown



OWL ANALOGY: ask before assuming

Staged plan — preserve the research gate

A

Verify literature

full texts, appendices, active disconfirmation

B

Settle design

key schema, labels, policy mapping, feasibility

C

Core benchmark

only if gate survives; human gold labels

D

Minimal simulator

validate deterministic event projection first

E

Decide scope

retain / narrow / reframe / abandon

Current gate: Pass 3 incomplete · novelty unresolved · implementation and formal drafting not authorized

THE STUDY IN ONE SENTENCE

When agents disagree about the same state key, when is it safe to agree, when should the system verify, and when should it preserve uncertainty?



FOX ANALOGY — question carefully

Measure it. Don’t assume the intelligent method is better.

Research proposal only · no result claimed · novelty and proof obligations remain open

A hypothetical AI-agent team shares one key

The labels are illustrative only: no real identity, capabilities, integrations, roles or performance are implied.

KEY service.demo-api.primary_db

Scope: demo-api • current release window • database enum • exactly one value

Grok bot	GPT dots	Claude duet	Harkpro	Wajo Fo	Wabi
hypothetical write	hypothetical write	hypothetical write	hypothetical write	hypothetical write	hypothetical write
PostgreSQL	**PostgreSQL**	**PostgreSQL**	**DynamoDB**	**DynamoDB**	**DynamoDB**

3 candidate writes

primary_db = PostgreSQL

3 candidate writes

primary_db = DynamoDB

Same declared key + scope + interval: equal values may be equivalent; the two values cannot both occupy a single-valued key.

Do not collapse four different questions

In the example, disagreement is not “solved” by counting agents or choosing the last arrival.

1 PROVENANCE

Six hypothetical labels
Each write stays attached to its source.

Who said which value?

2 RELATION

Same-key / same-scope
PostgreSQL vs DynamoDB = hard-conflict candidate.

Can both be true?

3 POLICY ACTION

VERIFY
Only with a named evidence source + deterministic rule.

What should happen?

4 RESULTING STATE

Evidence not supplied
→ UNRESOLVED; keep claims + provenance.

What gets published?

TURTLE ANALOGY: pause when evidence is missing — teaching analogy only, not a protocol component.

Key design points to get right before any build

These are open design questions, not claims that the design is already settled.

Same-key contract

Key, type, role, time window, cardinality

Layer separation

Provenance → relation label → action → output state

Verify rule

Allowed evidence, freshness, deterministic tie behavior

Fair comparison

Same inputs; classifier scores separate from policy outcomes

Safety first

False merges, false conflicts, abstention, calibration, cost

M.E. feasibility

Prioritize a small core; stage simulator/scaling

Likely questions

Why not newest-wins? Arrival order is not truth. • Does Verify guarantee correctness? No. • Is novelty established? No—Pass 3 is incomplete.

Monday 26 October — ready-to-present checklist

No review time or official KEC module titles were supplied; confirm those with the reviewer.

Before the review

- [ ] Open the PDF once; check slide order and diagrams on the presentation device.
- [ ] Keep PPTX + slide PDF together; keep report PDF as backup/reference.
- [ ] Rehearse the 90-second opener and the one-key scenario.
- [ ] Call the six names hypothetical labels only; assign no real capabilities.
- [ ] State the gate honestly: Pass 3 incomplete, novelty unresolved, no implementation.
- [ ] Ask which official modules/template and next milestone the college expects.

Open with this

“Same events reaching all replicas does not guarantee safe meaning. I’ll test how same-key disagreement should be classified and when to resolve, verify or preserve uncertainty.”

Ask for feedback

Is the key/scope precise?
Which work should be core vs staged?
What must pass before implementation?

Pass 3 evidence • close overlaps

- **Same-key commit and memory policy:** MemTX v2: evidence-bearing same-slot conflict gate; 90 cases / 5 backbones. Shared store, not async replica convergence.
- **Deferral is prior art:** TARL v5: append / noop / revise / reject_conflict / defer_verify; 5,422 cases with gold next state.
- **Classify then reconcile:** LatticeMind v1: Credibility vs Coordination; 73/75 on a 75-case ConflictBank set, one model.
- **Sources:** arXiv:2607.23929v2 · 2608.03699v5 · 2608.08236v1

Async delivery • measure what converges

- **Direct async example:** Governed Shared Memory v1: supersession 90/90 conditional on both writes landing; 98/200 overall.
- **Admission matters:** A synchronous dedup gate rejected 206/400 writes before the asynchronous detector; report gate and detector separately.
- **Different system questions:** CoAgent repairs plan invalidity; S-Bus rejects structural stale-read/write races; neither is same-key truth adjudication.
- **MELD caution:** Status convergence can coexist with order-sensitive live sets / retrieval: 112/200 orderings changed the live set.

Strong overlap ≠ exact experiment

- **StateFuse / TOKI:** Conflict-preserving replicated history, projection/abstention, provenance, time-aware policies and confirmation already appear.
- **StateFuse caveat:** Table 3 separates settings: conservative 64.9% accuracy / 100% conflict recall; aggressive 97.5% / 0% recall. Separate controlled 50-task loop: conservative 80%→100% after verification; not an async-convergence test.
- **SCF scope:** 600 simulated workflow runs; 27.9% precision / 65.2% recall / 39.0% F1 / 100% completion. Intent prevention ≠ claim convergence.
- **Survey / code benchmark:** Always-On: scoped 435-work survey, not exhaustive. CodeCRDT 5–10% = preliminary 60/600 manual sample.

Verification needs context

- **Evidence-pair errors:** ECon: weak-model recall limits; resolution may be unjustified or prior-informed.
- **Reference mismatch:** REF-NLI: 1,143 claim/evidence pairs, including ambiguous reference/context cases.
- **Adversarial NLI:** ANLI v2: ~19k / 47k / 103k examples; dev/test checked, training not fully verified.
- **Abstention is not verification:** Varshney et al. (17 datasets): no method consistently beats MaxProb across IID/OOD/ADV.

Gate stays open • no novelty verdict

- **Broad novelty is unsafe:** Classification, evidence-aware handling, defer/abstain, conflict preservation and async supersession each have close prior art.
- **Candidate comparison remains untested:** Same-key relation + evidence-based Verify vs fixed-winner vs preserve-unresolved under async reordering, with defined convergence outputs.
- **WH-1/2/3:** Exact authoritative definitions are missing; leave undefined and obtain the source wording.
- **Scope of this appendix:** Targeted excerpts, not full source/appendix audits or independent reproduction. Pack remains provisional; no implementation or convergence proof claimed.